Calculus

HW7

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Sec 1

Homework

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1) Find

a) $\int x^3 \cos x \, dx$ $u = x^3$ $dv = v' dx = \sin(x)\,dx$

$v' = \sin(x)$

+	x^3	$\sin(x)$
−	$3x^2$	$-\cos(x)$
+	$6x$	$-\sin(x)$
−	6	$\cos(x)$
+	0	$\sin(x)$

$$\int x^3 \cos x \, dx = x^3 \sin(x) + 3x^2 \cos(x) - 6x\sin(x) - 6\cos(x) + C$$

b) $\int x^5 \ln x \, dx$

$u = \ln x \qquad dv = x^5 dx$

$du = \frac{1}{x} dx \qquad v = \frac{x^6}{6}$

$$\int u\,dv = uv - \int v\,du$$
$$= \frac{x^6}{6}\ln x - \int \frac{x^6}{6} \cdot \frac{1}{x} dx$$
$$= \frac{x^6}{6}\ln x - \frac{1}{6}\int x^5 dx$$
$$= \frac{x^6}{6}\ln x - \frac{x^6}{36} + C$$

c) $\int e^x \sin x \, dx$

$u = \sin(x) \quad dv = e^x dx$

$du = \cos(x) dx \quad v = e^x$

$\int u \, dv = uv - \int v \, du$

$= e^x \sin(x) - \int e^x \cos(x) dx \rightarrow$

$u = \cos(x) \quad dv = e^x dx$

$du = -\sin(x) dx \quad v = e^x$

$\int e^x \cos(x) dx = e^x \cos(x) + \int e^x \sin(x) dx$

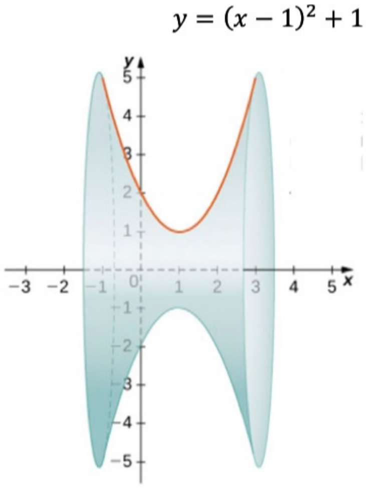
$\therefore \int e^x \sin(x) dx = e^x \sin(x) - \left(e^x \cos(x) + \int e^x \sin(x) dx\right)$

$= e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx$

$2 \int e^x \sin(x) dx = e^x \sin(x) - e^x \cos(x)$

$= \dfrac{e^x \sin(x) - e^x \cos(x)}{2} + C$

2) Find the volume of the solid generated by revolving the region bounded by $y = (x-1)^2 + 1$ and the x-axis on $[-1, 3]$ about the x-axis.
(using the **disk method**)



$x^2 - x - x + 1$

$$\text{Volume} = \pi \int_{-1}^{3} [R(x)]^2 \, dx$$

$$= \pi \int_{-1}^{3} [(x-1)^2 + 1]^2 \, dx$$

$$= \pi \int_{-1}^{3} [(x-1)^4 + 2(x-1)^2 + 1] \, dx$$

$$= \pi \left(\frac{(x-1)^5}{5} + \frac{2(x-1)^3}{3} + x \Big|_{-1}^{3} \right)$$

$$= \pi \left(\frac{2^5}{5} + \frac{2(2)^3}{3} + 3 - \left(\frac{(-2)^5}{5} + \frac{2(-2)^3}{3} - 1 \right) \right)$$

$$= \pi \left(\frac{32}{5} + \frac{16}{3} + 3 + \frac{32}{5} + \frac{16}{3} + 1 \right)$$

$$= \pi \left(\frac{64}{5} + \frac{32}{3} + 4 \right)$$

$$= \pi \left(\frac{192 + 160 + 60}{15} \right) = \frac{412\pi}{15}$$